

CHAPTER 13

Plant Growth and Development

Growth

- Which of the following statements is **not correct**? (2023-Manipur)
 - Phase of cell elongation of plant cells is characterized by increased vacuolation.
 - Cells in the meristematic phase of growth exhibit abundant plasmodesmatal connections.
 - Plant growth is generally determinate.
 - Plant growth is measurable.
- The process of growth is maximum during: (2020)
 - Lag phase
 - Senescence
 - Dormancy
 - Log phase
- Typical growth curve in plants is: (2015)
 - Stair-steps shaped
 - Parabolic
 - Sigmoid
 - Linear

Differentiation, Dedifferentiation, Redifferentiation and Development

- Formation of interfascicular cambium from fully developed parenchyma cells is an example for (2024)
 - Dedifferentiation
 - Maturation
 - Differentiation
 - Redifferentiation
- In tissue culture experiments, leaf mesophyll cells are put in a culture medium to form callus. This phenomenon may be called as: (2023)
 - Senescence
 - Differentiation
 - Dedifferentiation
 - Development
- The ability of plants to follow different pathways in response to environment leading to formation of different kinds of structures is called: (2022 Re)
 - Differentiation
 - Redifferentiation
 - Development
 - Plasticity
- The phenomenon by which the undividing parenchyma cells start to divide mitotically during plant tissue culture is called as: (2022 Re)
 - Secondary growth
 - Differentiation
 - Dedifferentiation
 - Redifferentiation

- Which one of the following plants does not show plasticity? (2022)
 - Maize
 - Cotton
 - Coriander
 - Buttercup
- Plants follow different pathways in response to environment or phase of life to form different kinds of structures. This ability is called: (2021)
 - Flexibility
 - Plasticity
 - Maturity
 - Elasticity

Plant Growth Regulators

- Match List-I with List-II: (2024 Re)

List-I		List-II	
(A)	Abscisic acid	(I)	Promotes female flowers in cucumber
(B)	Ethylene	(II)	Helps seeds to withstand desiccation
(C)	Gibberellin	(III)	Helps in nutrient mobilisation
(D)	Cytokinin	(IV)	Promotes bolting in beet, cabbage etc.

Choose the **correct** answer from the options given below:

- A-II, B-III, C-IV, D-I
 - A-III, B-II, C-I, D-IV
 - A-II, B-I, C-IV, D-III
 - A-II, B-I, C-III, D-IV
- F. Skoog observed that callus proliferated from the internodal segments of tobacco stem when auxin was supplied with one of the following except: (2024 Re)
 - Extract of Vascular tissues
 - Coconut milk
 - Abscisic acid
 - Yeast Extract
 - Given below are some statements about plant growth regulators. (2024 Re)
 - All GAs are acidic in nature.
 - Auxins are antagonists to GAs.
 - Zeatin was isolated from coconut milk.
 - Ethylene induces flowering in Mango.
 - Abscisic acid induces parthenocarpy.

Choose the correct set of statements from the options given below:

 - A, C, D
 - B, E
 - A, B, C
 - B, D, E

13. Auxin is used by gardeners to prepare weed free lawns. But no damage is caused to grass as auxin (2024)
 a. does not affect mature monocotyledonous plants.
 b. can help in cell division in grasses, to produce growth.
 c. promotes apical dominance.
 d. promotes abscission of mature leaves only.
14. Spraying sugarcane crop with which of the following plant growth regulators, increases the length of stem, thus, increasing the yield? (2024)
 a. Cytokinin b. Absciscic acid
 c. Auxin d. Gibberellin
15. Spraying of which of the following phytohormone on juvenile conifers helps in hastening the maturity period, that leads to early seed production? (2023)
 a. Absciscic Acid b. Indole-3-butyric Acid
 c. Gibberellic Acid d. Zeatin
16. Which hormone promotes internode/petiole elongation in deep water rice? (2023)
 a. 2, 4-D b. GA_3 c. Kinetin d. Ethylene
17. The phenomenon which is influenced by auxin and also played a major role in its discovery: (2023-Manipur)
 a. Phototropism b. Root initiation
 c. Gravitropism d. Apical Dominance
18. Match List-I with List-II. (2023-Manipur)

List-I		List-II	
(A)	Auxin	(I)	Promotes female flower formation in cucumber
(B)	Gibberellin	(II)	Overcoming apical dominance
(C)	Cytokinin	(III)	Increase in the length of grape stalks
(D)	Ethylene	(IV)	Promotes flowering in pineapple

Choose the **correct** answer from the options given below:

- a. A-II, B-I, C-IV, D-III b. A-IV, B-III, C-II, D-I
 c. A-I, B-III, C-IV, D-II d. A-III, B-II, C-I, D-IV
19. Which of the following growth regulators is an adenine derivative? (2022 Re)
 a. Absciscic acid b. Auxin
 c. Cytokinin d. Ethylene
20. Production of Cucumber has increased manifold in recent years. Application of which of the following phytohormones has resulted in this increased yield as the hormone is known to produce female flowers in the plants: (2022)
 a. Cytokinin b. ABA c. Gibberellin d. Ethylene
21. The gaseous plant growth regulator is used in plants to: (2022)
 a. Kill dicotyledonous weeds in the fields
 b. Speed up the malting process
 c. Promote root growth and root hair formation to increase the absorption surface
 d. Help overcome apical dominance

22. The plant hormone used to destroy weeds in a field is: (2021)
 a. NAA b. 2, 4-D c. IBA d. IAA
23. Name the plant growth regulator which upon spraying on sugarcane crop, increases the length of stem, thus increasing the yield of sugarcane crop. (2020)
 a. Gibberellin b. Ethylene
 c. Absciscic acid d. Cytokinin
24. Match the following concerning the activity/function and the phytohormone involved. (2020-Covid)

1.	Fruit ripener	(i)	Abciscic acid
2.	Herbicide	(ii)	GA_3
3.	Bolting agent	(iii)	2, 4-D
4.	Stress hormone	(iv)	Ethephon

Select the correct option from following:

- (1) (2) (3) (4)
 a. (iii) (iv) (ii) (i)
 b. (iv) (iii) (ii) (i)
 c. (iv) (ii) (i) (iii)
 d. (ii) (iii) (iv) (i)
25. Who coined the term 'Kinetin'? (2020-Covid)
 a. Darwin b. Went
 c. Kurosawa d. Skoog and Miller
26. It takes very long time for pineapple plants to produce flowers. Which combination of hormones can be applied to artificially induce flowering in pineapple plants throughout the year to increase yield? (2019)
 a. Auxin and Ethylene
 b. Gibberellin and Cytokinin
 c. Gibberellin and Absciscic acid
 d. Cytokinin and Absciscic acid
27. Fruit and leaf drop at early stages can be **prevented** by the application of: (2017-Delhi)
 a. Cytokinins b. Ethylene
 c. Auxins d. Gibberellic acid
28. You are given a tissue with its potential for differentiation in an artificial culture. Which of the following pairs of hormones would you add to the medium to secure shoots as well as roots? (2016 - II)
 a. Auxin and Absciscic acid
 b. Gibberellin and Absciscic acid
 c. IAA and Gibberellin
 d. Auxin and Cytokinin
29. The Avena curvature is used for bioassay of: (2016 -I)
 a. ABA b. GA c. IAA d. Ethylene
30. What causes a green plant exposed to the light on only one side, to bend toward the source of light as it grows? (2015)
 a. Light stimulates plant cells on the lighted side to grow faster.
 b. Auxin accumulates on the shaded side, stimulating greater cell elongation there.
 c. Green plants need light to perform photosynthesis.
 d. Green plants seek light because they are phototropic.

31. Auxin can be bioassay by: (2015 Re)
 a. Hydroponics
 b. Potometer
 c. Lettuce hypocotyl elongation
 d. Avena coleoptile curvature
32. Dr. F. Went noted that if coleoptile tips were removed and placed on agar for one hour, the agar would produce a bending when placed on one side of freshly cut coleoptile stumps. Of what significance is this experiment? (2014)
 a. It demonstrated polar movement of auxins
 b. It made possible the isolation and exact identification of auxin
 c. It is the basis for quantitative determination of small amounts of growth-promoting substances
 d. It supports the hypothesis that IAA is auxin
33. Which one of the following growth regulators is known as 'stress hormone'? (2014)
 a. Indole acetic acid
 b. Absciscic acid
 c. Ethylene
 d. GA₃
34. During seed germination its stored food is mobilised by: (2013)
 a. Gibberellin
 b. Ethylene
 c. Cytokinin
 d. ABA

Answer Key

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|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (d) | 3. (c) | 4. (a) | 5. (c) | 6. (d) | 7. (c) | 8. (a) | 9. (b) | 10. (c) |
| 11. (c) | 12. (a) | 13. (a) | 14. (d) | 15. (c) | 16. (d) | 17. (a) | 18. (b) | 19. (c) | 20. (d) |
| 21. (c) | 22. (b) | 23. (a) | 24. (b) | 25. (d) | 26. (a) | 27. (c) | 28. (d) | 29. (c) | 30. (b) |
| 31. (d) | 32. (b) | 33. (b) | 34. (a) | | | | | | |